

Independent reanalysis of the largest sporadic Creutzfeldt–Jakob disease genome-wide association study confirms all published loci and refines the STX6 signal

Arthur Araújo da Silva — Independent Researcher, Brazil *Projeto DCJ - Lito (independent data-organization initiative)* · Preprint draft v0.3 — 2026-08-29 — target: *bioRxiv*

Abstract

Background. Sporadic Creutzfeldt–Jakob disease (sCJD) is the most common human prion disease. The largest genome-wide association study (GWAS) of sCJD (GCST90001389; 4,110 cases, 13,569 controls; *Lancet Neurology* 2020) reported three genome-wide significant loci at PRNP, STX6 and GAL3ST1. Independent replication of summary statistics by parties outside the original consortium remains rare.

Methods. We downloaded the publicly deposited summary statistics (6,314,492 variants, GRCh37) and performed an end-to-end independent quality control and replication analysis using only open-source tooling implemented from first principles in Python's standard library: genomic-control inflation (λ), genome-wide significance screening, locus annotation against Ensembl GRCh37 coordinates, Wakefield approximate Bayes factors, linkage-disequilibrium clustering against the 1000 Genomes phase 3 panel via the Ensembl REST API, and stratified λ diagnostics by minor-allele-frequency bin.

Results. (1) We independently reproduce **all three published loci**: PRNP chr20:4,672,307 ($p = 1.62 \times 10^{-15}$), GAL3ST1 chr22:30,950,360 ($p = 6.18 \times 10^{-10}$) and STX6 chr1:180,961,245 ($p = 7.51 \times 10^{-9}$); 41 variants reach $p < 5 \times 10^{-8}$; no malformed records were found; global $\lambda = 1.0587$. (2) Fine-mapping with real LD shows the STX6 signal is a single cohesive haplotype block: the lead cluster ($r^2 \geq 0.80$ with anchor rs11586493) carries **90.5% of regional posterior mass**; at PRNP, 100% of posterior falls within a single linkage block around codon 129 (rs1799990 appears among proxies, $r^2 \approx 0.68$); the GAL3ST1 signal is poorly tagged by common-panel proxies (max $r^2 \approx 0.45$), consistent with a lower-frequency haplotype. Notably, **rs3747957 — the index variant independently identified by an Oxford multi-omic study in *Brain* (2025) — is present in the 2020 sumstats with $p = 9.74 \times 10^{-9}$ and identical effect direction ($\beta = -0.148$)**, ranked 11th regionally, demonstrating that the 2020 data already contained evidence later validated functionally. (3) Stratified λ analysis across MAF bins spans only 0.016 across minor-allele-frequency strata (<0.05 : $\lambda=1.062$; ≥ 0.45 : $\lambda=1.070$), arguing against substantial residual population stratification. (4) As a parallel verification exercise on expression data, we reproduce exactly the neuronal-loss/gliosis transcriptional signature of GSE160208 (184 differentially expressed genes under the original authors' criteria; $r = 1.000$ for top-ranked genes), and show that the blood microRNA signature of GSE140069 (Nat Commun 2020) loses formal significance after standard adjustment for age, sex and RIN (84 miRNAs significant at $FDR < 0.05$ without covariate adjustment $\rightarrow 1$ after adjustment), although directionality and nominal significance of the four discovery miRNAs persist ($p = 7 \times 10^{-4} - 4 \times 10^{-2}$).

(5) A five-cohort brain eQTL meta-analysis (~2,182 samples) shows the STX6 lead variant is a genome-wide significant brain eQTL ($p = 6.6 \times 10^{-47}$) with 89% effect-direction concordance and complete regional posterior concentration on both-signal hypotheses ($H2+H3+H4 = 1.000$ across all parametrizations; $H3+H4 \approx 0.995$ under our default prior; the standard R `coloc.abf` implementation with its own prior assigns 98% to $H4$), supporting expression-mediated mechanism; a splicing (sQTL) scan of the same five cohorts finds **no significant cis-sQTL for STX6 in any of them** (0 clusters; the GWAS signal also does not colocalize with any neighbouring-gene splicing signal), making expression the only detectable molecular trait at the locus; and (6) a blind whole-genome clumping scan rediscovers all three published loci as the top three clusters (3/3), demonstrating pipeline sensitivity rather than circular confirmation.

Conclusions. Public prion-disease summary statistics support full independent replication by non-consortium parties. We document a previously unreported consistency between the 2020 sCJD GWAS and the 2025 functional index variant rs3747957, provide cluster-level credible sets for all three loci, and quantify how covariate confounding can inflate blood-based biomarker signatures. All code, checksums and reports are openly available.

1. Introduction

Human prion diseases are fatal neurodegenerative conditions caused by misfolding of the cellular prion protein (PrP^{Sc}). Sporadic CJD accounts for ~85% of cases. Beyond the PRNP codon 129 modifier, host genetic modifiers were established by the 2020 GWAS of Mead et al., which identified PRNP, STX6 and GAL3ST1 at genome-wide significance.

Independent reanalysis serves three purposes that the original study cannot: (i) it verifies that public deposition is complete and internally consistent; (ii) it stress-tests conclusions with methods chosen independently of the original pipeline; (iii) it lowers the barrier for citizen-science participation in rare-disease research, following the precedent of Minikel & Vallabh.

Here we report a fully independent, from-scratch reanalysis of GCST90001389 and two companion expression datasets (GSE160208 brain tissue; GSE140069 whole blood), executed without access to individual-level genotypes and with all statistics implemented from first principles.

2. Data

Resource	Accession	Content
sCJD GWAS sumstats	GCST90001389	6,314,492 variants, GRCh37, β /SE/p/EAF
Brain expression	GSE160208	47 samples (27 CJD / 20 controls), NanoString 800 genes
Blood microRNA	GSE140069	57 sCJD / 48 controls, 939 miRNAs
LD reference	1000 Genomes phase 3 (ALL)	via Ensembl REST ld endpoint

Integrity: SHA-style MD5 checksums recorded before analysis (Appendix of project repository).

3. Methods

3.1 GWAS quality control

Streaming parse of the compressed sumstats (single pass, constant memory). Malformed-line count, allele-frequency sanity, χ^2 statistic per variant as $(\beta/\text{SE})^2$, genomic-control factor $\lambda = \text{median}(\chi^2)/0.454936$ computed globally and within pre-registered MAF strata (<0.05; 0.05–0.25; 0.25–0.45; ≥ 0.45). Genome-wide threshold $p < 5 \times 10^{-8}$.

3.2 Locus annotation

GRCh37 gene coordinates retrieved live from Ensembl REST (never from memory) for PRNP, STX6, GAL3ST1 windows (± 50 kb around lead).

3.3 Approximate-Bayes-factor fine-mapping

Per-variant Wakefield ABF with prior variance $W = 0.04$ on $\log(\text{OR})$: $\text{ABF}_i = \sqrt{(\text{SE}^2/(\text{SE}^2+W)) \cdot \exp(\chi^2_{iW}/(2(\text{SE}^2+W)))}$. Regional posteriors $\pi_i \propto \text{ABF}_i$. Because individual genotypes are unavailable, we report **cluster-level credible mass**: variants are grouped by pairwise $r^2 \geq 0.80$ with the regional lead using Ensembl REST LD (1000G phase 3 ALL); posterior mass of the lead cluster versus remainder quantifies whether the signal is one haplotype block or dispersed. This is explicitly an approximation — joint models (SuSiE/FINEMAP) require genotypes.

3.4 Expression analyses

GSE160208: Welch t-tests on log2-transformed normalized intensities, frontal-cortex-only contrasts, Benjamini–Hochberg FDR; exact replication of the original authors' criteria ($p < 0.05 \wedge |\log_2 \text{FC}| > 1$) alongside our FDR < 0.05 criterion. GSE140069: log2(x+1) transform; primary model = OLS $\log_2 \sim \text{group} + \text{sex} + \text{age} + \text{RIN}$ (covariates from the series matrix; the original publication adjusted age); sensitivity analyses unadjusted and detection-filtered. Effect sizes as Cohen's d. All statistics cross-validated against R anchors (t distribution CDF and BH procedure matched to $\leq 10^{-13}$ relative error; permutation calibration of FDR).

3.5 Colocalization with brain eQTLs

For the STX6 region (chr1:180.9–181.1 Mb GRCh37), we tested whether the GWAS signal and STX6 expression quantitative trait loci (eQTLs) share a causal variant, using the regional approximate-Bayes-factor colocalization framework (Giambartolomei et al. 2014) implemented from the published equations: prior $W = 0.04^2$; priors $p_1 = p_2 = 10^{-4}$, $p_{12} = 10^{-5}$ (sensitivity: 10× more conservative). eQTL summary statistics were retrieved as remote tabix region queries (pysam/htslib) from the eQTL Catalogue release 8, which uniformly reprocessed GTEx v10 and independent brain cohorts: CommonMind DLPFC (n=586), ROSMAP DLPFC (n=560), BrainSeq DLPFC (n=479), GTEx v10 DLPFC (n=285) and GTEx v10 cerebellum (n=272). eQTL coordinates (GRCh38) were converted to GRCh37 with a single Ensembl-mapped offset (–30,864 bp), and alleles harmonised to the ALT/effect convention (0 incompatible variants discarded). Because no single cohort is individually powered, we additionally performed an inverse-variance-weighted meta-eQTL across the five cohorts (~2,182 brain samples), yielding a lead-cluster eQTL signal of $p = 6.6 \times 10^{-47}$. Implementation validation: (i) an independent reimplement of the ABF/posterior equations in R agrees with our Python code to 6 decimal places ($H_3 = 0.994997$ vs 0.9950); (ii) the standard R `coloc.abf` (coloc 5.2.3) reproduces the combined both-signal posterior ($H_2 + H_3 + H_4 = 1.000$) with per-variant PP.H4 Spearman $\rho = 0.91$ against our ABF-diagonal mass — while demonstrating that the shared-vs-distinct split (H_4 vs H_3/H_2) is prior-dependent, as declared. LD-panel sensitivity: re-running the fine-mapping with the population-matched 1000 Genomes EUR panel leaves the STX6 lead-cluster posterior unchanged (90.5% at $r^2 \geq 0.8$), improves PRNP tagging (58.9% → 80.9%), and leaves GAL3ST1 poorly tagged in both panels (consistent with a lower-frequency haplotype). Splicing scan: leafcutter sQTL summary statistics for the same five cohorts (eQTL Catalogue r8 cc files = significance-passed cis pairs per intron cluster; cluster identity normalised by genomic coordinates) were harmonised to the GWAS and tested by colocalization per cohort×cluster; clusters were attributed to genes by the catalogue's `gene_id` column (STX6 = ENSG00000135823, GRCh37 chr1:180,941,861–180,992,047, verified via Ensembl REST lookup).

3.6 Blind whole-genome clumping (discovery sensitivity)

To demonstrate that our pipeline would find the known loci rather than merely confirm them, we scanned all 6,314,492 variants blind (threshold $p < 10^{-5}$) and applied greedy distance clumping (± 500 kb; lead = smallest p within cluster), with no locus annotation used until after cluster definition. Known loci were matched post hoc.

4. Results

4.1 Sumstats integrity and inflation

Zero malformed records among 6,314,492 lines. Global $\lambda_{\text{GC}} = 1.0587$ (liminal, acceptable). Stratified λ :

MAF stratum	n	λ
<0.05	325,236	1.0617
0.05–0.25	532,664	1.0579
0.25–0.45	328,920	1.0547
≥ 0.45	76,078	1.0703

Gradient 0.0156 — inconsistent with major residual population stratification, which preferentially inflates common variants.

4.2 Independent replication of all three loci

Locus	Our best hit (GRCh37)	p	β	Annotation
PRNP	chr20:4,672,307 C>T	1.62×10^{-15}	-0.219	PRNP region; rs60704301/rs2093390/rs4254562
GAL3ST1	chr22:30,950,360 T>C	6.18×10^{-10}	-0.169	GAL3ST1 promoter/5' region
STX6	chr1:180,961,245 G>A	7.51×10^{-9}	-0.149	intragenic STX6 (Ensembl 180,941,861-180,992,047)

All 41 genome-wide-significant variants fall within these three regions; none elsewhere. A blind whole-genome clumping scan (Section 3.6) independently rediscovers all three loci as the top three clusters, confirming pipeline sensitivity rather than circular confirmation.

4.3 Cluster-level fine-mapping

- **PRNP**: 337 regional variants; anchor rs60704301 (merged $\rightarrow$ rs2093390). Posterior mass 100% within $r^2 \geq 0.50$ of the anchor in **both LD panels** (58.9% at $r^2 \geq 0.80$ with the ALL panel; **80.9%** with the population-matched EUR panel); codon-129 variant rs1799990 appears among proxy rsIDs. The entire signal is one haplotype structure around the prion-protein gene. - **STX6**: 162 variants; anchor = lead rs11586493; 20/20 top variants panel-covered, max $r^2 = 1.00$; **90.5%** of mass in the $r^2 \geq 0.80$ lead cluster — a single cohesive block that includes rs3747957 ($r^2 = 0.99$); **identical (90.5%) under the EUR panel**. - **GAL3ST1**: 322 variants; lead absent from 1000G phase 3 (anchor rs386462923 $\rightarrow$ rs8142452, rank 3); only 4/20 top variants panel-covered; best cross- $r^2 \approx 0.45$ — **unchanged under the EUR panel**. The signal is poorly tagged by common proxies — consistent with a lower-frequency haplotype and an explicit caveat for imputation-based replication.

4.4 Consistency with the 2025 functional index variant

The Oxford multi-omic study (*Brain*, 2025) nominated synonymous/missense variant **rs3747957** (chr1:180,953,853 GRCh37) as the STX6 index. In the 2020 sumstats this variant has $p = 9.74 \times 10^{-9}$, $\beta = -0.148$ — same direction as our lead ($\beta = -0.149$) — ranking 11th of 162 regional variants. The 2020 dataset therefore already carried the association later validated functionally; we supply the explicit numerical bridge.

4.5 Expression signatures replicate exactly (brain) but are fragile under covariate adjustment (blood)

Brain GSE160208: 437/800 genes FDR<0.05; 184 DEGs under original criteria — identical count and rank order ($r = 1.000$ top-10). Blood GSE140069: unadjusted log2-Welch yields 84 significant miRNAs (10 $\uparrow$ /74 $\downarrow$); OLS adjusting age+sex+RIN leaves **1** (hsa-miR-500a-3p); the four discovery miRNAs retain direction and nominal significance ($p = 7 \times 10^{-4}$ – 4×10^{-2}) but only hsa-miR-93-5p survives FDR within the detection-filtered universe ($q = 0.048$). Cases were on average 12.8 years older than controls (66.4 vs 53.6) with lower RNA integrity (RIN 5.59 vs 6.50) — a textbook confounding structure that the original paper partially addressed (age via Partek GSA) and that fully explains the discrepancy between naive and adjusted counts. Furthermore, a novel cross-compartment integration analysis (not performed by either original study) shows that experimentally validated targets of the four discovery blood miRNAs are NOT over-represented among the up-regulated brain DEGs (hypergeometric, all $q = 1.0$; overlaps at or below chance), arguing that the blood signature reflects peripheral processes rather than the cerebral transcriptional program — cautioning against naive blood $\rightarrow$ brain causal inference.

4.7 Colocalization: the STX6 GWAS signal sits inside a significant brain eQTL block

Per-cohort colocalization is power-limited (eQTL $n \leq 586$; the GWAS lead is a suggestive single-cohort eQTL with concordant direction in 78–81% of regional variants). The five-cohort meta-eQTL, however, renders the lead GWAS variant a **genome-wide significant STX6 eQTL** ($p = 6.6 \times 10^{-47}$; $z \approx 14$) with **89% effect-direction concordance** across 390 harmonised variants. In the regional colocalization framework, H_0+H_1 collapse to ≈ 0 and the entire posterior sits on both-signal hypotheses — **$H_2+H_3+H_4 = 1.000$ in every parametrization tested** ($H_3+H_4 \approx 0.995$ under our default prior; the standard `R coLoc.abf` 5.2.3 with its own prior assigns 98% to H_4): both signals are real and confined to the same LD block. Our implementation was validated by an exact reimplement of the ABF/posterior equations in R (agreement to 6 decimal places) and by the standard `coLoc.abf` package (per-variant PP.H4 Spearman $\rho = 0.91$ against our ABF-diagonal mass) — which also demonstrates that the shared-vs-distinct split (H_4 vs H_3/H_2) is prior-dependent under $r^2 \geq 0.97$ LD, as declared. The strict split between H_4 (shared causal variant) and H_3 (distinct causal variants) is formally unidentifiable here because all cluster members are in $r^2 \geq 0.97$ — a documented limitation of colocalization under strong LD. Given (i) the significant meta-eQTL at the GWAS lead, (ii) direction concordance, and (iii) our fine-mapping showing a single posterior cluster, the totality of evidence supports regulation of STX6 expression as a plausible mechanism of the association, consistent with the independent functional nomination of rs3747957 by the Oxford multi-omic study.

4.8 Splicing: no significant cis-sQTL for STX6 in any brain cohort

The eQTL Catalogue release 8 provides leafcutter splicing-QTL summary statistics for the same five brain cohorts. We scanned every significant intron-cluster (the `cc` files contain only significance-passed cis pairs) whose cis window intersects the STX6 region, harmonised variants to the GWAS and ran the validated colocalization per cohort $\times$ cluster (6 tests total). **Zero significant cis-sQTL clusters for STX6 exist in any cohort** — in contrast to the strong expression QTL ($p = 6.6 \times 10^{-47}$) — and no neighbouring-gene splicing signal (KIAA1614, QSOX1, RP5-1180C10.2) colocalizes with the GWAS either (all PP.H4 ≈ 0 ; Bonferroni-aware interpretation). At this locus, expression is the only detectable molecular trait; a splicing-mediated component is neither supported nor formally excluded (absence of evidence in significance-filtered files is not evidence of absence), but there is no positive splicing signal to interpret.

4.9 Blind clumping rediscovers all published loci

The blind scan produced 35 independent clusters; the top three are exactly PRNP (chr20:4,672,307; $p = 1.62 \times 10^{-15}$), GAL3ST1 (chr22:30,950,360; $p = 6.18 \times 10^{-10}$) and STX6 (chr1:180,961,245; $p = 7.51 \times 10^{-9}$) — **3/3 known loci rediscovered without any locus hint**, in correct significance rank. The strongest residual cluster (chr16:15,539,902; $p = 5.73 \times 10^{-8}$) is borderline genome-wide significant and merits independent follow-up; we report it as an observation, not a novel-locus claim.

4.10 Ethics and scope

No individual-level human data were generated or obtained beyond public deposits; no patient-identifiable information was processed. This work is a verification contribution and makes no clinical claims.

5. Discussion

Our results deliver the three things an independent reanalysis can uniquely provide. First, **verification**: every number in the deposited sumstats parsed cleanly, all 41 genome-wide-significant variants sit exactly where the original consortium reported them, and genomic inflation is liminal and uniform across allele-frequency strata — the public record of the largest sCJD GWAS is trustworthy, and we publish the checksums and code to let anyone re-verify this in minutes.

Second, **the STX6 association is expression-mediated to the limit that summary statistics allow**. A five-cohort brain eQTL meta-analysis makes the GWAS lead variant a genome-wide significant eQTL of STX6 ($p = 6.6 \times 10^{-47}$) with concordant effect direction in 89% of regional variants, and regional colocalization concentrates the entire posterior on shared-block hypotheses. Strict single-variant attribution (H_4) is unidentifiable under $r^2 \geq 0.97$ — we say so explicitly — but the combination of significant meta-eQTL, direction concordance and a single fine-mapping cluster is exactly the pattern expected if the 2020 association and the

2025 functional nomination of rs3747957 are two views of the same regulatory event.

Third, **a previously unreported numerical bridge between two landmark studies**. The 2025 Oxford multi-omic study functionally nominated rs3747957 as the STX6 index variant. We show the 2020 sumstats already contained that association at $p = 9.74 \times 10^{-9}$ with matching effect direction — evidence that was present but unremarked for five years. This is a concrete demonstration of why open summary statistics matter: discoveries sometimes sleep in deposited data until someone looks.

Fourth, **a cautionary quantification for biomarker research**. The blood microRNA signature for sCJD (Nat Commun 2020) collapses from 84 $FDR < 0.05$ -significant miRNAs (no covariate adjustment) to 1 after standard covariate adjustment, because cases were on average 12.8 years older than controls. Directionality and nominal significance of the four discovery miRNAs survive — consistent with a genuine but weaker-than-presented signal. We stress this is not an accusation: the original authors adjusted age themselves; our contribution is making the magnitude of naive-vs-adjusted divergence explicit and reproducible for future biomarker pipelines.

Methodologically, we demonstrate that a complete GWAS quality-control, fine-mapping and colocalization pipeline can run with a standard-library statistical core (all inferential statistics implemented from first principles and cross-validated against R anchors), relying on external open-source libraries only for figure rendering (matplotlib) and remote tabix access to eQTL summary statistics (pysam/htslib). This lowers the entry barrier for researchers in resource-limited settings, including in countries like Brazil where sCJD surveillance exists but prion-genetics capacity is thin.

Finally, our cluster-level credible analysis shows the STX6 signal is one cohesive LD block whose posterior mass concentrates on the lead haplotype rather than dispersing across independent false positives. Formal joint fine-mapping with individual genotypes remains the gold standard; we offer ours as the honest ceiling achievable from summary statistics alone.

6. Limitations

- Summary-statistics-only: no conditional/joint fine-mapping; cluster-level inference only. - Expression datasets lack individual-level covariates for GSE160208 (no age/PMI metadata). - Our OLS implementation cannot reproduce Partek's gene-specific variance correction; differences in Section 4.5 may partly reflect estimator choice. - Colocalization under $r^2 \geq 0.97$ cannot split H3 (distinct variants) from H4 (shared); we report block-level support plus direction concordance instead of an H4 point estimate. - Brain eQTL cohorts (DLPFC/cerebellum, adult) only approximate the affected tissue and cell types in sCJD; cell-type-specific (e.g. microglial) eQTL may differ. - Single-author independent initiative; peer review pending (bioRxiv DOI upon submission).

7. Data & code availability

Sumstats GCST90001389 (GWAS Catalog); GEO GSE160208/GSE140069. Full pipeline (pure-Python stdlib), reports with every intermediate number, figure-generation scripts and MD5 checksums: github.com/BlackYuriJDU/dcj-lito (release v1.0.0-preprint; archived on Zenodo, DOI 10.5281/zenodo.22164910).

8. References

1. Mead S. et al. *Lancet Neurol* 2020;19:793–802 (PMID 32949544). 2. Areskeviciute A., Litman T. et al. *Int J Mol Sci* 2020;22:140 (PMID 33375642). 3. Norsworthy P.J. et al. *Nat Commun* 2020;11:3960 (PMID 32769986). 4. Multi-omic STX6 study. *Brain* 2025 (rs3747957 index). 5. Minikel E.V. et al. *Sci Transl Med* 2016;8:340ra73. 6. Wakefield J. *Genet Epidemiol* 2009;33:79–86 (ABF). 7. Benjamini & Hochberg. *JRSS-B* 1995;57:289–300. 8. Vallabh & Minikel — Prion Alliance / cureffi.org (open-science precedent).

Figures: volcano_gse160208.png, volcano_gse140069.png, heatmap_top_genes.png (pipeline/reports/figuras/). Fine-mapping numbers from relatorio_finemap_loci.md v2 and relatorio_lambda_gc.md. All statistics cross-checked against R anchors; adversarial audit report included in the repository

(colaboracao/laudo_estatistico_adversarial.md).